CATALYTIC CONVERTER [SKYACTIV-G 1.3, SKYACTIV-G 1.5]

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Purpose, Function

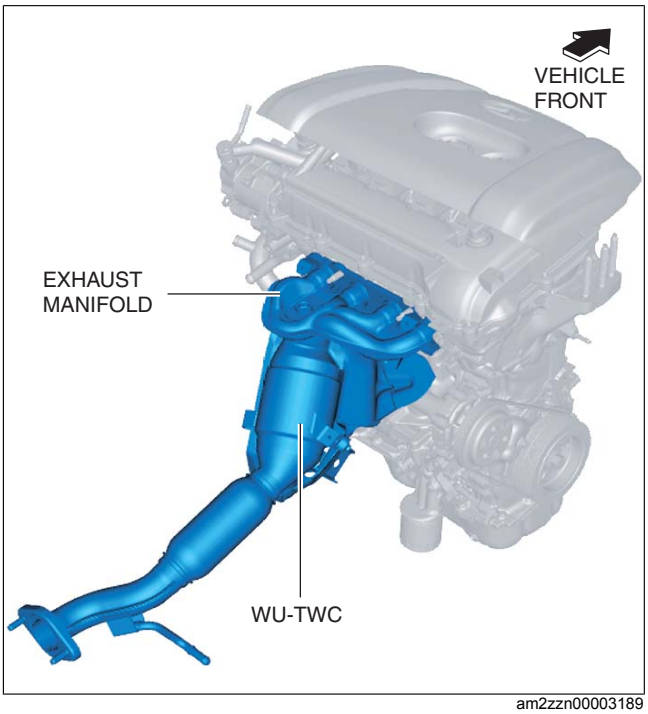
- The catalytic converter purifies contaminants in the exhaust gas by utilizing the chemical reactions of the three-way catalytic converter.
- A WU-TWC and a TWC have been adopted.

Construction

WU-TWC

- The catalytic converter (WU-TWC) is located behind the engine.

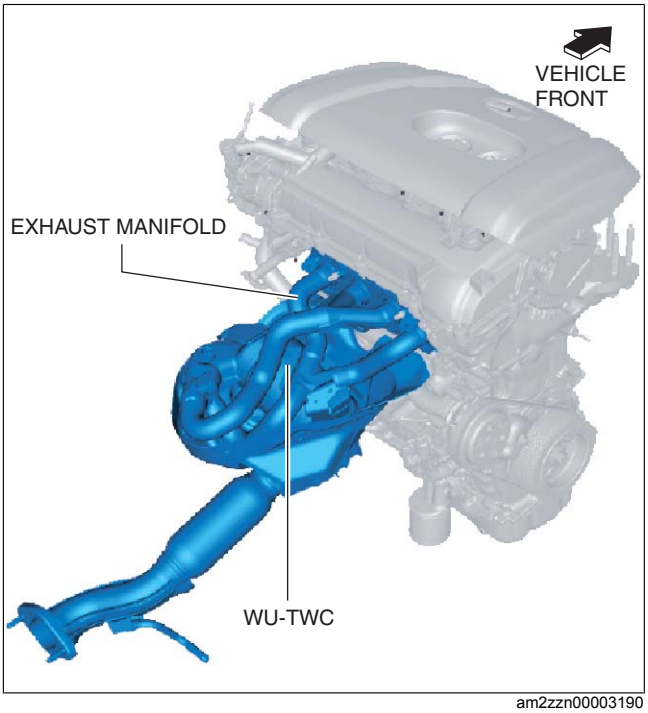
SKYACTIV-G 1.3, SKYACTIV-G 1.5 (with 4-1 exhaust system)



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SKYACTIV-G 1.5 (with 4-2-1 exhaust system)

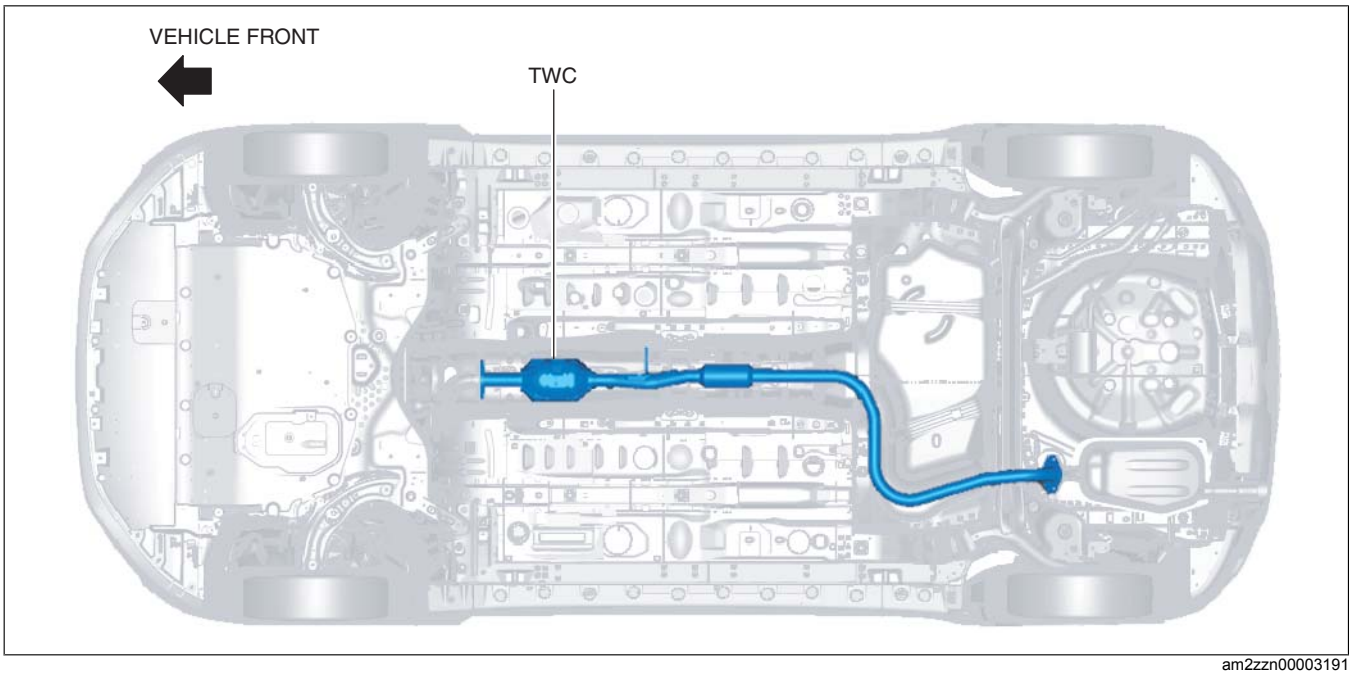
- The WU-TWC is integrated with the exhaust manifold.



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TWC

- The catalytic converter (TWC) is located behind the catalytic converter (WU-TWC).



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Operation

- Contaminants in the exhaust gas (HC, CO, NOx) are purified by oxidization and deoxidization while they are passing through the catalytic converter.
  - — Oxidization process
    - Noxious hydrocarbon (HC) and carbon monoxide (CO) are bound with oxygen and converted to non-noxious carbon dioxide and water.
      $O_2+HC+CO \rightarrow CO_2+H_2O$
  - — Deoxidization process
    - Noxious nitrogen oxide (NOx) is converted to non-noxious nitrogen and oxygen. A part of the oxygen generated in this process is used in the oxidization process.
      $NOx \rightarrow N_2+O_2$